1. Consider the finches data set from past assignments. Predict the **'winglength'** variable based on the categorical **'island'** variable, run ANOVA on the data, and describe your analysis. We've already worked with this data set a few times, so you do not need to explain the variable and data set's origin, but you must include at least the following steps in your write up:

   (a) Construct a boxplot to display the data with respect to your analysis.

   (b) List the assumptions that apply to the ANOVA test. Do these assumptions appear to be met, based on your plot? Why or why not?

   (c) State your null and alternative hypotheses to test whether winglength differs by island.

   (d) List all code to produce a linear model and the corresponding ANOVA table.

   (e) Use the ANOVA table to make a formal conclusion in both statistical terms (do we reject H0?) and in non-statistical terms (what does this mean in the context of our data?). Use $\alpha = 0.05$ for your conclusions.

   (f) Confirm that ANOVA's coefficients match the group means, showing your work through R functions (**HINT**: consider which island is acting as the intercept).

   (g) Give an interpretation of the residual for the **tenth** observation in this dataset in context. What sort of difference does it represent? State whether the observed value is **more** or **less** than predicted.

   NOTE: fitted(), coefficients(), and residuals() will be helpful for the final two parts. As a reminder, you can also load the data using:
   `finches <- read.csv("https://raw.githubusercontent.com/njlytal/MATH314/main/finches.csv")`

2. Load the `diamonds` data set, found within the ggplot2 package, using the code `diamonds <- ggplot2::diamonds`. Consider how the response variable **price** can be predicted by the categorical variable **cut**, which represents the quality of cut for the diamonds.

   (a) Create a boxplot that compares the prices for the different quality cuts of diamond.

   (b) Say you want to test multiple pairwise comparisons at a combined confidence level of $\alpha = 0.05$. How many such pairwise comparisons exist for the different qualities of cut, and what will your adjusted confidence level α^* be for each comparison?

   (c) Filter the data to include only the "Good" and "Very Good" cut qualities. Use t.test() to perform a pairwise comparison at the adjusted confidence level α^*, and conclude whether the two group mean prices are different.

   (d) Repeat step C, but this time compare the "Fair" and "Premium" cut qualities and report your conclusion.